Programming Languages

CPCS301

CS21 First Semester Final Exam

Unsolved version

Question1. Choses the correct answer, one answer for each question

1. Programming Domains that must be fast execution and with low level feature

a) Scientific applications
b) Web Software
c) Artificial intelligence
d) Systems programming

2. Type checking is the activity Which help in

a) Writability
b) Reliability
c) Readability
d) Cost

3. Symbols, consisting of names rather than numbers, are manipulated

a) Artificial intelligence
b) Scientific applications
c) Systems programming
d) Web Software

4. The ……. And ……. let the variables be accessible

a) Readability and life time
b) Life time and writability
c) Scope and life time
d) Scope and writability

5. Used in Web documents to specify the layout of information

a) Markup
b) Functional
c) Imperative
d) Object-oriented

6. Used to describe the static semantics of a language

a) Attribute Grammars
b) Context-Free Grammar
c) Backus-Naur Form
d) Grammar

7. Which of the following data types cannot used as array index

a) Boolean
b) Integer
c) Chat
d) Float

8. In perl, %salaries = (“Gary" => 75000, “Perry" => 57000, “Mary” => 55750,“Cedric” => 47850); is

a) Associative arrays
b) Arrays
c) Record
d) Union

9. What is the result of float (9/4)

a) 2.25
b) 2
c) 3.0
d) 2.0

Question 2. Match the definition with the term.

1. Array	**a)** Binding of subscript ranges and storage allocation is dynamic and can change any number of times
2. Dangling Pointer	**b)** Binding of subscript ranges and storage allocation is dynamic and cannot change the size
3. Record	**c)** Is a heterogeneous aggregate of data elements
4. Heap Dynamic Array	**d)** Disallowed in Ada Language
5. Fixed Heap Dynamic Array	**e)** Is a homogeneous aggregate of data elements
6. Memory leakage	**f)** Allocated heap-dynamic variable that is no longer accessible to the user program

Question 3. Determine whether the following sentences are **TRUE (T)** or **FALS (F)**

1. **( )** Selection statement provides the means of choosing between two between two path only
2. **( )** Control expression in C switch statement can be only an integer type
3. **( )** Multiple entries are possible only in languages that include goto and statement labels
4. **( )** In Iteration statement if the step size is absence the default value = 0
5. **( )** Loops can be nested in conditions but conditions cannot be nested in loops
6. **( )** In general, indentation has no effect on semantics in contemporary languages & therefore ignored by their compilers
7. **( )** Counter-Controlled Loops has pre and post test
8. **( )** Iteration is controlled throw a Boolean excretion and counter

Question 4.

a) Convert this Java code to let else match the *<u>outer if</u>*

```
If (sum == 0)
        If (count == 0){
                Result = 0;
        }
Else {
        Result = 1;
}
```

b) **Convert this Ruby code to let else match the *inner if***

If (sum == 0) then
    If (count == 0) then
        Result = 0;
    end
Else
    Result = 1;
 	
end	

Question 5. What the value of x after evaluate the code and answers the following question

```
Fun (int *i){
     i += 2;
     j *= i;
     return 6;
}

Main {
     Int x = 1,  y = 2;
     x += 3;
     x = x + fun (&x);
}
```

a) **Is there a side effect if the statement is evaluated from <u>left to right (Yes or No)</u>**

b) **Is there a side effect if the statement is evaluated from <u>right to left (Yes or No)</u>**

Question 6.

```
x = 50; y = 0;
If ((x > y) || x--/5)
```

a) **Is there a short circuit evaluation? <u>(yes or no) with explanation</u>**

b) **When y = 100, is there a side effect? <u>(yes or no)</u>**

Question 7. Assume the following calling sequence and that **<u>static scoping is used,</u>** what variables are visible and hidden during execution of the last subprogram activated? Include with each visible variable the name of the unit where it is declared.

```
Procedure main is
     X,Y,Z : integer;
     Procedure initializer() is
          X,Y : integer;
          Procedure convert() is
          Z : integer;
          begin
          …   -----------------------------> A
          end;
     begin
     …     --------------------------------> B
     end;
     Procedure Compute() is
          Z,Y : integer;
          Procedure Evaluate() is
          begin
          …
          end;
     begin
     …    ---------------------------------> C
     end;
begin
…     -------------------------------------> D
end;
```

Points	Visible variables
A	
B	
C	
D	

Question 8. Assume the following calling sequence and that **dynamic scoping is used,** what variables are visible and hidden during execution of the last subprogram activated? Include with each visible variable the name of the unit where it is declared.

Main calls Sub3, Sub3 call Sub2, Sub2 calls Sub1

```
Main(){
Int x,y,z;
Sub1(){
Int Z;
}

Sub2(){
Int Y,Z;
}

Sub3(){
Int X;
}
}
```

Points	Visible variables	Hidden variables
main	X, Y, Z from Main	All other variables in Sub1, Sub2 and Sub3
Sub3		
Sub2		
Sub1		

Question 9. Evaluate the given C code.

```
Sum(int a, int b, int c = 10){
     x = a + b + c;
}
```

a) **Sum(7,1)**

b) **Sum(4,2,7)**

Question 10. This code written in imaginary language, trace the program and fill the table with the appreciate values.

```
Main:
     Int I = 1, j = 2;
     Fun ( i , j );
     Print (i , j);

Fun (int k, int m):
     k = k + 1;
     m = m + I;
     Print (i , j, k , m);
```

	Semantic model	Print(i,j,k,m) in subprogram				Print(i,j) in main	
		i	j	k	m	i	j
Bass by reference							
Bass by value							

Note: We do not remember the exact code in question 7,8 but you got the idea

Thanks and kisses goes to CS21